FRONTEND ENGINEER · RÉSUMÉ 2026

Hart Luo 骆皓天

Frontend Engineer · UNSW (QS 19) BSc & MIT · Focus: *AI agents, canvas & complex UI state*

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00 · ABOUT Highlights

01 Core frontend engineer for DreamFace's multimodal Agent (**100k+ DAU**) and infinite canvas (**20k DAU**), owning stream recovery, interactive task continuation, output consistency and incremental persistence.

02 Author of the open-source canvas engine infinite-canvas-core, optimised against benchmarks: **48 µs** viewport queries over 100,000 nodes, worst-case frame 44 → **13.8 ms**.

03 UNSW Computer Science and IT graduate; nearly five years studying and interning in Australia; fluent working English.

01 · WORK Experience

2026.04 – Present

Hithink RoyalFlush — Frontend Engineer

DreamFace · overseas AIGC product with **300k+ DAU** across iOS, Android and web; own the core Agent and Canvas frontend, with contributions to mobile web and the admin console

Vue 3 / Nuxt 3 / TypeScript / SSE / Canvas / Vitest

Multimodal Agent & infinite canvas

core developer · Agent 100k+ DAU · Canvas 20k DAU

— **Recoverable sessions:** designed session state and multi-session orchestration shared by desktop, mobile web and Canvas. Modelled running, suspended and terminal states; implemented cursor-based reconnection, history/live reconciliation, ID migration and replay merging for background generation and re-entry recovery.

— **Interactive continuation:** built clarification forms, voice selection, credit confirmation and top-up continuation. Reused submission IDs on retries, checked receipts after lost responses and reconciled card state to handle duplicate clicks, uncertain submissions and stale events while resuming the original task.

— **Agent-to-canvas consistency:** mapped placeholders, final media and server snapshots by output ID; coordinated local saves with remote reconciliation, filled missing nodes and removed stale placeholders. Scoped delayed callbacks to project sessions to handle duplicate placement and cross-project writes.

— **Document persistence:** designed CanvasDocument, incremental commands and undo / redo, replacing localStorage with server persistence. Built debounced merging, idempotent retries, revision-conflict retry and weighted batching, fixing failed saves on **300**-node bulk edits.

— **Reliability:** helped launch the canvas from zero; fixed **5** classes of renderer/document divergence, including ghost nodes, selection snap-back, delete crashes, offline freezes and bulk-asset failures. Added recovery, continuation and project-isolation regression cases; documented and presented the architecture.

Promotions and analytics

— Shipped site-wide banners and media campaign pop-ups across the web app and admin console; owned core event tracking, resolving missing triggers and conflicting IDs before release.

02 · INTERN Internships

2025.06 – 2025.11

DVINE Experiences — Software / IT Intern · Australian tourism company

Booking and operations automation on the GoHighLevel low-code platform

GoHighLevel / Forms / Workflows

— Built configuration-driven booking forms on GHL Forms / Custom Fields and trigger → action workflows covering booking creation, confirmations, status updates and reminders; shipped **10+** forms and modules.

2024.06
2024.09

Beyondsoft Frontend Intern · on site with an Alibaba project team

Supply-chain admin system (enterprise SaaS) delivered for ChaPanda; owned the transport management and order visualisation modules

React / Umi / Ant Design Pro / ECharts

— Shipped **10+** pages with Umi, React, TypeScript and Ant Design Pro, **10+** reusable components and **20+** ECharts reports, from requirements through release.

03 · BUILD Projects

2026.09
Open source

infinite-canvas-core Headless infinite-canvas engine

github ↗

General canvas capabilities rebuilt in my own time as a standalone open-source engine (no company code)

TypeScript / Canvas2D / Vitest / Playwright

— **Spatial index**: a uniform grid with cell coordinates packed exactly into one float64 and oversized items kept apart — **48 µs** viewport queries over 100,000 nodes (604 µs for a linear scan, **13×**).

— **Rendering**: batching and a sub-pixel LOD buffer cut the worst 100k-node frame from 44 to **13.8 ms**; benchmarks caught a **3.3×** regression from downlevelled `#private` fields, now test-guarded.

— **Architecture**: scale-and-translate camera, geometric hit testing, patch-based undo and snapping; DOM-free core runs in a Worker or Node; **160+** unit tests.

2026.08
Open source

Hand Review Trainer Offline-first PWA

live ↗

A full training loop: play → record → EV analysis → flag mistakes → review → long-term statistics

React / TypeScript / IndexedDB / Playwright

— Rules engine, 6 AI strategies, per-decision EV review and 15 mistake categories; state is separated from UI and all randomness comes from a seed, so any session replays exactly.

— **985** automated tests: 100,000 cases cross-checked against a reference, invariants over 10,000 self-play sessions, Playwright on desktop and mobile; works offline.

At UNSW
Startup

UNSW HandbookX Frontend lead · course search and planning

github ↗

Built because the official course handbook was slow to use; 100+ student users during campus testing

Vue 3 / TypeScript / NestJS

— Set the team's coding conventions and component style guide, ran weekly code reviews, and built shared components and utilities to cut duplicated work.

04 · STUDY Education

2024.09
2026.01

UNSW Sydney Master of Information Technology · GPA 6.12 / 7.0 · QS 19

2021.09
2024.06

UNSW Sydney Bachelor of Science (Computer Science) · GPA 6.03 / 7.0 · QS 19

05 · CRAFT Skills

FRONTEND	Proficient in **Vue 3**, **TypeScript** and JavaScript; Nuxt 3, **React**, Umi, Vite, Ant Design, TailwindCSS, Zustand, Sass, ECharts, PWA / IndexedDB
AI AGENT	SSE sessions, cursor-based recovery, interactive task continuation, history/live reconciliation, multimodal output integration with canvas
GRAPHICS	Canvas2D and complex interaction: coordinate transforms, spatial indexing, viewport culling, incremental rendering; document models, incremental saving and undo / redo design
ENGINEERING	Vitest / Playwright, GitHub Actions CI, Sonar; benchmarking, profiling and analytics; Claude Code / Codex for development and review
OTHER	Node.js, WebSocket, RESTful API design; data structures and algorithms
LANGUAGE	Mandarin (native), English (professional working proficiency)